

Which step(s) of the viral life cycle would most likely occur in lytic phages?

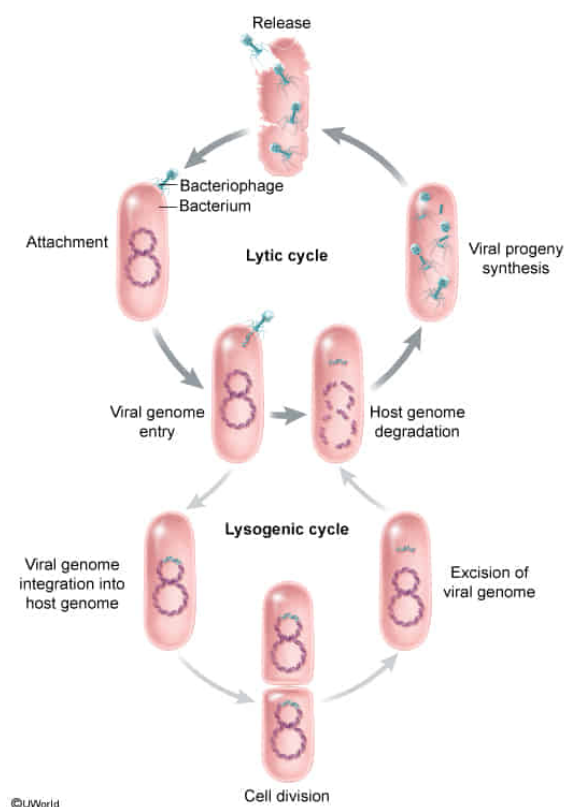
- I. Injection of viral genetic material into the bacterial host
- II. Integration of viral DNA into the bacterial genome
- III. Degradation of the host genome

- ☐ A. I only [9%]
- ☐ B. III only [11%]
- ☐ C. I and II only [25%]
- ✓ ☒ D. I and III only [53%]

Correct

53% answered correctly

Explanation:



All **viruses** are obligate **intracellular parasites** because they cannot **replicate** independently outside of a host cell. **Bacteriophages** (sometimes shortened to phages) are viruses that exclusively infect bacteria. The life cycle of a particular bacteriophage can be lytic or lysogenic, which determines the method by which the phage uses the host's cellular machinery to create new progeny. The general steps involved during the **lytic life cycle** are described as follows:

1. **Attachment:** The bacteriophage contacts the bacterial cell wall and attaches to the host bacterium using its tail fibers.
2. **Viral genome entry:** The phage uses its tail sheath to inject its genome into the

cytoplasm of the bacterial host **(Number I)**.

3. **Host genome degradation:** Viral enzymes degrade the host genome into its nucleotide components to provide the building blocks for replication of the viral genome **(Number III)**.
4. **Synthesis:** Loss of the bacterial (ie, host) chromosome ends the synthesis of host molecules, now under the control of the viral genome. As a result, the host machinery (eg, ribosomes) begins to synthesize the components needed for new viral progeny, which then assemble inside the host cell.
5. **Release:** Many newly assembled viral progeny (virions) are released as the bacterium disintegrates (lysis) due to the action of lysozymes on the host cell wall.

Therefore, the steps of a viral life cycle that would most likely occur in lytic phages are **injection of viral genetic material into the bacterial host** and **degradation of the host genome**.

(Number II) In contrast, bacteriophages that have a *lysogenic* life cycle also attach to the bacterium but **integrate** their genome **into the host DNA**; these viruses are called prophages. This viral genome integration enables the host cell to survive and divide normally. While the host cell divides, the integrated viral DNA is **replicated** with the host DNA, and associated viral repressor proteins prevent the viral genome from being transcribed (latency). Induction can occur in response to environmental factors (eg, chemicals, radiation), causing the viral genome to excise itself from the host chromosome and enter the lytic life cycle.

Educational objective:

Bacteriophages use the host cell's machinery and resources to replicate their genome and synthesize viral proteins to form new virions. Phages with a lytic life cycle replicate rapidly and release progeny via lysis of the host cell. In contrast, phages with a lysogenic life cycle integrate their genome with the host genome and replicate as the cell divides.

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